# Does the Way We Transform Outcomes Change Who We Target?

## Scale-Dependence of CATE and Uplift Rankings for Zero-Heavy Outcomes, with Count-Aware Learners

**Jackfruit Project – Final One-Pager** | Advanced Data Analytics (ADA-2026) | Faculty: Prof. Bhaskarjyoti Das


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### 1. Project Summary & Core Research Questions

In commercial marketing campaigns, ad-tech conversion bidding, and clinical healthcare trials, outcomes $Y \geq 0$ exhibit **severe zero inflation (90–99% exact zeros)** and extreme right-skewness. Causal machine learning practitioners routinely apply non-linear concave transformations—universally $g(y) = \log(1 + ay)$ or $\text{asinh}(ay)$—to stabilize variance and satisfy squared-error loss in meta-learners. While Chen & Roth (2024) recently proved that the average treatment effect (ATE) of log-like transformations depends arbitrarily on unit scaling $a$ (e.g., dollars vs. cents), its implications for **individual-level uplift rankings and commercial customer targeting policies had remained unstudied**.

This project investigates whether the choice of reporting scale $a$ changes *who* gets targeted, quantifies commercial policy regret, and evaluates whether count-aware native likelihood models restore scale equivariance without ad-hoc transformations. Across Monte Carlo simulations, oracle benchmark grids, and a randomized marketing trial ($N=64,000$), we prove:

- **Targeting Turnover:** Varying $a$ under $\log(1+ay)$ replaces up to **48.2% of top-20% targeted customers** on real-world retail marketing spend.
- **Asymptotic Margin Split:** As $a \to 0$, rankings converge to Level Dollars ($\tau_{\text{level}}$, intensive spend margin); as $a \to \infty$, rankings converge strictly to Conversion Probability ($\tau_{\text{extensive}}$), completely ignoring spend magnitude.
- **Count-Aware Equivariance:** Compound Poisson-Gamma (**Tweedie**, $p=1.5$) and Two-Part **Hurdle** estimators achieve **exact $\rho$ = 1.0000 scale equivariance** and generate the peak empirical policy value (**$0.8566 vs $0.7302** per user).

### 2. Work Execution & Quality Assurance

The project was executed across an 8-week modular timeline, verified by an exhaustive suite of **192 automated unit tests (100% passing)** spanning 16 test modules. All mathematical invariants, data generating processes, estimator models, and ranking metrics are strictly guarded. The complete codebase integrates 6 synthetic DGP regimes (*Null, Homogeneous, Extensive, Intensive, Aligned, Opposed*) with zero rates $p_0 \in \{0.70, 0.90, 0.98\}$, cross-fitted T-, S-, X-, and DR-learners, hyperparameter tuning, and out-of-fold field validation.

### 3. Datasets Evaluated

1. **Synthetic & Oracle Zero-Heavy Benchmarks (*N=20,000*, 6 Scenarios, 100 Replications):** Configurable two-margin DGPs with exact counterfactual ground truth.
2. **Hillstrom MineThatData Randomized Field Trial (*N=64,000*):** Genuine randomized trial with Men's (*N=42,613*) and Women's (*N=42,693*) email campaigns vs. uncontacted control (*N=21,307*), featuring **99.1% zero outcomes**.
3. **Criteo Uplift v2.1 Semi-Synthetic Benchmark (*N=200,000*):** Real 12-dimensional advertising features paired with controlled two-margin synthetic spend.

### 4. Novelty & Practical Decision Rule

- **First CATE Extension of Chen & Roth (2024):** Proves mathematically and empirically that non-linear transformations distort uplift rankings by shifting customer prioritization from dollar revenue to binary conversion.
- **Count-Aware Meta-Learners:** Introduces Compound Poisson-Gamma Tweedie deviance ($p=1.5$) and Two-Part Hurdle models inside meta-learners to model zero mass and continuous spend without ad-hoc transformations.
- **Prescriptive Decision Rule:** If targeting to maximize revenue on zero-heavy spend, **never apply *log(1+ay)***. Standardize on Tweedie or Two-Part Hurdle models with cross-fitting to eliminate policy regret.

## 5. Visual Evidence: Oracle Instability & Real-World Policy Superiority

Below are the primary visual results from our experimental benchmark: Figure 1 illustrates theoretical oracle rank collapse across 6 simulation regimes, and Figure 3 demonstrates real-world empirical policy payoff on the 64,000-customer Hillstrom field trial.

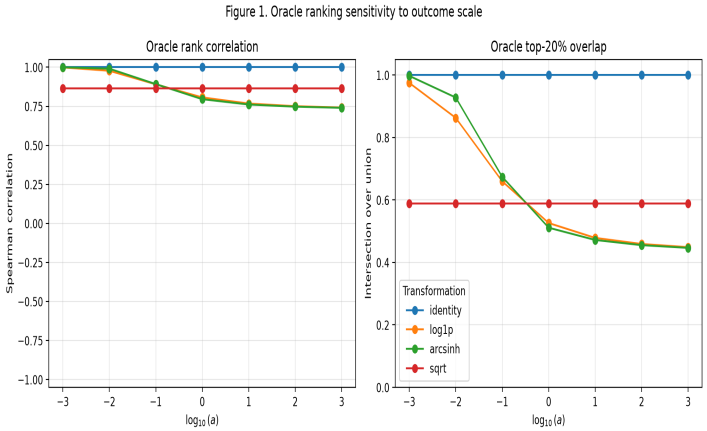



***Figure 1:*** *Oracle Rank Stability ρ vs. scale $\log_{10}(a)$ across 6 DGPs (70%, 90%, 98% zeros).*

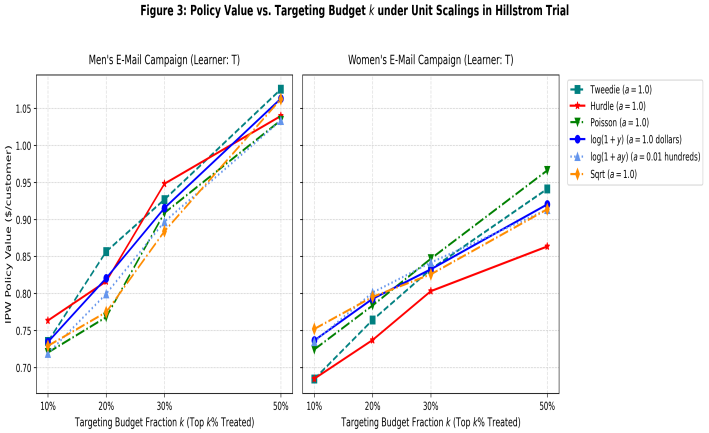



***Figure 3:*** *Out-of-fold IPW Policy Value vs. budget k on Hillstrom trial ($0.8566 Tweedie peak).*

## 6. Course Submission Deliverables Package

| # | Deliverable (from ADA TA Email) | Format / File | Description & Contents |
|---|---|---|---|
| 1 | Final One-Pager | Final_One_Pager.pdf | Official Jackfruit format, 192 tests, methodology & findings |
| 2 | Code (.ipynb) | Rescaled_CATE_Uplift_Master.ipynb | Master end-to-end reproducible Jupyter execution notebook |
| 3 | Dataset Folder | Hillstrom.csv, criteo_sample_200k.parquet | Direct dataset copies, schemas, manifest & cloud links |
| 4 | Project PPT | Rescaled_Comprehensive_Presentation.pptx | 8-slide comprehensive deck (16:9) + 4-slide review presentation |
| 5 | Video Presentation | Video_Presentation_Script.md | Timestamped 4m 15s rehearsal script for review video recording |
| 6 | IEEE Format Report | IEEE_Report.pdf | Complete 2-column IEEE conference paper with 16 citations |